Qassim University

College of Computer

Information Technology Department

University Event Management System (UEMS)

Course Title: IT Project Management

Course Code: IT 612

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EXECUTIVE SUMMARY

This document presents a comprehensive Project Management Plan for the development of the University Event Management System (UEMS). The UEMS is designed to automate and centralize the management of academic and non-academic events, such as conferences and workshops, at the university. Currently, these processes are handled manually, leading to significant inefficiencies, scheduling conflicts, and budget mismanagement.

By applying the 10 Knowledge Areas of the Project Management Body of Knowledge (PMBOK), this plan outlines the strategy for a 10-week project lifecycle. Key deliverables include a robust Project Charter, a detailed Work Breakdown Structure (WBS), a comprehensive Risk Register, and a strategic Communication Plan. While this document does not involve the actual coding of the system, it provides the professional framework necessary to ensure the project is completed on time, within budget, and to the satisfaction of all university stakeholders.

1. INTRODUCTION

1.1 Project Overview

The University Event Management System (UEMS) is a strategic IT initiative aimed at modernizing the university's approach to event organization. In the current academic environment, events are a core component of knowledge exchange. However, the lack of a digital platform has resulted in fragmented data and operational bottlenecks. The UEMS will serve as a unified web-based solution to manage every phase of an event, from initial planning and room reservation to participant registration and post-event reporting.

1.2 Problem Statement

The manual coordination of events at the university has led to three primary challenges:

1. **Operational Inefficiency:** Manual registration and tracking are time-consuming and prone to human error.
2. **Resource Conflicts:** Without a centralized calendar, multiple events are often booked for the same venue or time slot.
3. **Financial Lack of Transparency:** Budgeting and expense tracking are currently decentralized, making it difficult for management to assess the financial health of individual events.

1.3 Project Goals and Objectives

The primary goal of this project management plan is to guide the development of the UEMS through a structured 10-week lifecycle. Specific objectives include:

- **Centralization:** Consolidate all event-related data into a single, accessible database.
- **Automation:** Automate the registration, notification, and certification processes.
- **Optimization:** Implement a real-time resource allocation system to eliminate scheduling conflicts.

1.4 Document Purpose

This Project Management Plan (PMP) acts as the primary reference for the project team and stakeholders. It defines the "how, when, and who" of the project, ensuring that the development team has a clear roadmap to follow and that the university management has a benchmark against which to measure progress.

2. PROJECT CHARTER

Project Name: University Event Management System (UEMS)

Project Manager: Ajmal Hamidi

Project Sponsor: Qassim University IT Department / University Management

Start Date: February 16, 2026

End Date: April 30, 2026

2.1 Business Case

The university currently organizes numerous academic and non-academic events, including conferences and workshops . These activities are vital for the university's reputation and knowledge dissemination. However, the existing manual registration and coordination processes are inefficient and prone to errors. Transitioning to the UEMS will centralize data, improve participant tracking, and enhance the overall professional image of university events . The investment in this system is justified by the expected reduction in administrative overhead and the elimination of costly scheduling conflicts .

2.2 Problem Statement

The current manual event management process at the university has resulted in:

- **Communication Gaps:** Information is siloed, leading to poor coordination between departments.
- **Scheduling Conflicts:** Overlapping venue bookings and time-slot clashes occur frequently due to the lack of a centralized calendar.
- **Financial Mismanagement:** Budget tracking for events is inconsistent, making it difficult to monitor expenses effectively.
- **Inefficient Tracking:** Participant registration and post-event data collection are slow and lack accuracy.

2.3 Project Objectives (SMART)

The primary objectives for the UEMS project are:

- **Specific:** Develop a comprehensive Project Management Plan for a web-based event management system to automate registration and resource allocation.
- **Measurable:** Achieve a 100% reduction in venue booking overlaps upon system implementation.

- **Achievable:** Complete the planning and management framework within the university's existing IT academic guidelines.
- **Relevant:** Align with the university's digital transformation goals to modernize administrative workflows.
- **Time-bound:** Finalize the complete project management report and presentation by April 30, 2026.

2.4 High-Level Requirements

The UEMS must meet the following high-level technical and functional requirements:

- **Centralized Dashboard:** A unified interface for administrators to view all upcoming events and resource statuses.
- **Online Registration:** A secure portal for participants to register for conferences and workshops.
- **Resource Management:** A module to manage venue availability and equipment allocation.
- **Reporting Tool:** Automated generation of participant lists and budget summaries .

2.5 Key Stakeholders

- **Project Sponsor:** University IT Management.
- **Project Manager:** Ajmal Hamidi.
- **System Users:** Faculty members, administrative staff, and event organizers.
- **End Users:** Students and external participants (attendees).
- **Technical Team:** Database administrators and web developers.

2.6 Assumptions and Constraints

- **Assumptions:**
    - Stakeholders will provide necessary requirements within the first two weeks.
    - The university's existing server infrastructure is sufficient to host the planned system.
- **Constraints:**
    - **Time:** The project must be completed within a strict 10-week academic window.
    - **Budget:** The project must adhere to the allocated university budget for IT initiatives.

- **Scope:** The project is limited to the management and planning phases; actual coding is not required for this submission .

2.7 Project Manager Authority

The Project Manager, Ajmal Hamidi, is authorized to:

- Lead the project planning and team coordination.
- Allocate resources for the development of project management artifacts (WBS, Gantt charts, etc.).
- Communicate directly with university stakeholders regarding project status and requirements.

3. PROJECT LIFE CYCLE MODEL

3.1 Selection: Agile (Scrum) Methodology

For the development of the University Event Management System (UEMS), the **Agile (Scrum)** life cycle model has been selected. Unlike the traditional Waterfall model, which is rigid and linear, Agile allows for iterative development, frequent feedback, and the ability to adapt to changing requirements—a common scenario in university environments where multiple departments (IT, Faculty, Administration) are involved.

3.2 Justification for the Agile Model

The choice of an Agile Life Cycle is based on the following factors:

- **Stakeholder Feedback:** Transitioning from a manual to a digital system requires constant input from university staff to ensure the user interface (UI) and functionality meet their actual needs.
- **Risk Mitigation:** By developing the system in "Sprints," the project team can identify technical issues or integration problems early in the process rather than at the end of the project.
- **Flexibility:** As the university's event requirements evolve, Agile provides the flexibility to reprioritize features (e.g., prioritizing "Registration" over "Digital Certification") based on immediate needs.

3.3 UEMS Project Phases

The life cycle for the UEMS project is divided into the following structured phases over the 10-week duration:

1. **Initiation & Analysis (Weeks 1-2):**
   - Definition of project goals and signing of the Project Charter.
   - Gathering detailed requirements from the IT Department and Event Organizers.
2. **Design & Sprint Planning (Weeks 3-4):**
   - Creating the system architecture and database schema.
   - Drafting the User Interface (UI) wireframes.
   - Breaking down the Product Backlog into manageable tasks.
3. **Iterative Development (Sprints) (Weeks 5-8):**
   - **Sprint 1:** Core registration module and database connectivity.
   - **Sprint 2:** Resource allocation (venue booking) and calendar integration.

- **Sprint 3:** Reporting tools, payment gateway, and notification system.

4. **Testing and Quality Assurance (Week 9):**
   - Conducting User Acceptance Testing (UAT) with university staff.
   - Fixing bugs and optimizing system performance.
5. **Deployment & Closing (Week 10):**
   - Final system handover and documentation.
   - Review of "Lessons Learned" and official project closure.

3.4 Project Governance

To ensure the life cycle remains on track, a **Daily Stand-up** (simulated) and **Sprint Reviews** will be conducted. This ensures that any "Scheduling Conflicts" or "Communication Gaps" mentioned in the Project Brief are addressed in real-time through the structured Agile framework.

4. INTEGRATION MANAGEMENT PLAN

4.1 Overview of Integration Management

The Integration Management Plan for the University Event Management System (UEMS) serves as the "glue" that holds all other project management areas together. Its primary purpose is to coordinate the various processes—such as scope, schedule, cost, and quality—to ensure the project objectives are met within the 10-week timeframe. As the Project Manager, Ajmal Hamidi will ensure that all project artifacts are synchronized and that any changes in one area (e.g., a delay in schedule) are reflected and managed in others (e.g., budget or quality).

4.2 Project Management Plan Development

The UEMS Project Management Plan is a consolidated document that integrates all subsidiary plans (Scope, Schedule, Cost, Risk, etc.).

- **Expert Judgment:** We will leverage the expertise of the Qassim University IT Department and academic event organizers to ensure the plan is realistic and technically sound.
- **Meetings:** Weekly synchronization meetings will be held with the project team to ensure all integrated components are aligned.

4.3 Directing and Managing Project Work

During the execution phase (Weeks 5-8), the focus shifts to performing the activities defined in the project plan.

- **Project Management Information System (PMIS):** We will use tools like MS Project and Excel to track task completion and resource utilization.
- **Deliverables:** The primary deliverable of this phase is the functional UEMS software modules (Registration, Resource Allocation, and Reporting).

4.4 Monitoring and Controlling Project Work

To ensure the project stays on track, continuous monitoring is required.

- **Performance Reporting:** Monthly status reports will compare actual progress against the **Project Baseline**.
- **Issue Log:** Any technical or administrative hurdles (such as server downtime or stakeholder unavailability) will be documented in an issue log with assigned owners and resolution dates.

4.5 Integrated Change Control

IT projects frequently face "Scope Creep." To prevent this, a formal Integrated Change Control process is established:

1. **Change Request:** Any stakeholder requesting a change to the UEMS (e.g., adding a new payment gateway) must submit a formal Change Request Form.
2. **Impact Analysis:** The project team evaluates the impact on time, cost, and risk.
3. **Change Control Board (CCB):** A small committee (Project Manager and IT Sponsor) will review and either Approve, Reject, or Defer the change.
4. **Update Baseline:** If approved, the project plan and baselines are updated to reflect the new scope.
5. **Configuration Management:** We will use version control (e.g., Git) to manage the technical specifications and documentation of the UEMS, ensuring that only approved versions of the system are deployed.

4.6 Project Closure

In the final week (Week 10), the project will undergo formal closure:

- **Administrative Closure:** Finalizing all documentation, archiving project records, and releasing resources.
- **Technical Closure:** Handing over the UEMS system to the university's permanent IT support team.
- **Lessons Learned:** Conducting a post-project review to document what went well and what could be improved for future university IT initiatives.

5. SCOPE MANAGEMENT PLAN

5.1 Introduction to Scope Management

The Scope Management Plan for the University Event Management System (UEMS) ensures that the project includes all the work required, and only the work required, to complete the project successfully. Given the potential for "Scope Creep" in university software projects—where various departments may request additional features mid-project—this plan establishes a rigorous boundary for the development team.

5.2 Project Scope Statement

The UEMS project focuses on delivering a centralized digital platform for managing university events.

- **Project Inclusions (In-Scope):**
  - Development of a web-based portal for participant registration and payment.
  - Creation of a centralized venue and equipment booking calendar.
  - Implementation of an automated notification system (Email/SMS) for event updates.
  - A reporting module for generating attendance lists and financial summaries.
  - Integration with the university's existing student/staff authentication system.
- **Project Exclusions (Out-of-Scope):**
  - Physical event organization (e.g., hiring catering or cleaning services).
  - Marketing and promotional activities for specific events.
  - Procurement of end-user hardware (e.g., laptops or tablets for registration desks).
  - Mobile application development (initial release is web-only).

5.3 Work Breakdown Structure (WBS)

The WBS is a deliverable-oriented hierarchical decomposition of the work to be executed by the project team. It breaks down the UEMS project into manageable "Work Packages."

WBS Code	Level 1: Project	Level 2: Phase/Major Deliverable	Level 3: Work Package
1.0	**UEMS Project**	**1.1 Project Management**	1.1.1 Project Charter & Planning Documents
			1.1.2 Status Reports & Stakeholder Meetings
		1.2 Requirements & Analysis	1.2.1 Stakeholder Interview Results
			1.2.2 Technical Requirements Specification
		1.3 System Design	1.3.1 Database Schema & ER Diagram
			1.3.2 UI/UX Wireframes & Prototyping
		1.4 Development (Sprints)	1.4.1 Registration & Payment Module
			1.4.2 Resource Allocation & Calendar
			1.4.3 Admin Dashboard & Reporting
		1.5 Testing & QA	1.5.1 Unit & Integration Testing
			1.5.2 User Acceptance Testing (UAT)
		1.6 Deployment & Closeout	1.6.1 Training Materials & User Manuals
			1.6.2 Final Handover & Lessons Learned

5.4 WBS Dictionary (Overview)

Each work package identified in the WBS will have a corresponding entry in the WBS Dictionary. This ensures that the team understands the specific requirements for each task. For example:

- **Work Package 1.4.1 (Registration Module):** This includes the frontend forms for data entry, backend validation logic, and secure integration with the payment gateway. Success is measured by the successful creation of a participant record in the database upon payment.

5.5 Scope Validation and Control

To prevent unauthorized changes, scope validation will occur at the end of each Sprint.

- **Validation:** The Project Manager and University IT Sponsor will review deliverables against the requirements defined in Section 5.2.
- **Control:** Any deviations from the WBS must go through the formal Integrated Change Control process defined in Section 4.5. This ensures that any new feature requests (e.g., adding a "Live Streaming" module) are evaluated for their impact on the 10-week deadline and the project budget.

6. SCHEDULE MANAGEMENT PLAN

6.1 Introduction to Schedule Management

The Schedule Management Plan for the University Event Management System (UEMS) establishes the criteria and the activities for developing, monitoring, and controlling the project schedule. Given the fixed 10-week academic window (ending April 30, 2026), time is the most significant constraint for this project. This plan ensures that all tasks identified in the WBS are sequenced logically to meet the final deadline.

6.2 Scheduling Methodology

The project utilizes a **Hybrid Scheduling Approach**:

- **Predictive (CPM):** The Critical Path Method is used for the Initiation, Planning, and Closing phases to ensure foundational tasks are completed on time.
- **Iterative (Agile):** The Execution phase is managed through three distinct Sprints, allowing for flexibility in feature development.

6.3 Activity List, Durations, and Dependencies

Below is the master schedule data. This table can be used to generate the Gantt Chart in MS Project or Excel.

ID	Task Description	Duration	Predecessor	Start Date	End Date
1	**Phase 1: Initiation & Analysis**	**2 Weeks**	-	Feb 16	Feb 27
1.1	Project Charter Approval	3 Days	-	Feb 16	Feb 18
1.2	Stakeholder Interviews	4 Days	1.1	Feb 19	Feb 24
1.3	Technical Requirements Finalization	3 Days	1.2	Feb 25	Feb 27
2	**Phase 2: System Design**	**2 Weeks**	**1.3**	Mar 2	Mar 13
2.1	Database Schema Design (ERD)	5 Days	1.3	Mar 2	Mar 6
2.2	UI/UX Wireframing	5 Days	2.1	Mar 9	Mar 13
3	**Phase 3: Development (Sprints)**	**4 Weeks**	**2.2**	Mar 16	Apr 10
3.1	Sprint 1: Registration & Payment	10 Days	2.2	Mar 16	Mar 27
3.2	Sprint 2: Resource Allocation	5 Days	3.1	Mar 30	Apr 3
3.3	Sprint 3: Reporting & Notifications	5 Days	3.2	Apr 6	Apr 10
4	**Phase 4: Testing & QA**	**1 Week**	**3.3**	Apr 13	Apr 17
4.1	System Integration & Security Test	3 Days	3.3	Apr 13	Apr 15
4.2	User Acceptance Testing (UAT)	2 Days	4.1	Apr 16	Apr 17
5	**Phase 5: Final Review & Closing**	**2 Weeks**	**4.2**	Apr 20	Apr 30
5.1	Revision & Final Report Compilation	5 Days	4.2	Apr 20	Apr 24
5.2	Final Presentation & Handover	4 Days	5.1	Apr 27	Apr 30

6.4 Project Milestones

Milestones are significant points in the project that represent the completion of a major deliverable.

- **M1:** Project Charter Signed (End of Week 1)
- **M2:** Design Approved (End of Week 4)
- **M3:** Development Completed (End of Week 8)

- **M4:** UAT Approval (End of Week 9)
- **M5:** Final Submission (April 30, 2026)

6.5 Critical Path Analysis

The **Critical Path** for the UEMS project consists of the sequence of activities that must be completed on schedule to ensure the project finishes by April 30. Any delay in these tasks will delay the entire project.

- **Path:** Charter -> Requirements -> Database Design -> UI Wireframing -> Sprint 1 -> Sprint 2 -> Sprint 3 -> UAT -> Final Report.
- **Total Float:** Because of the strict 10-week limit, the project has "Zero Float" on the critical path.

6.6 Schedule Control

To manage the schedule effectively:

- **Progress Tracking:** Actual vs. Planned progress will be reviewed every Thursday.
- **Schedule Compression:** If the project falls behind, "Crashing" (adding more resources to Critical Path tasks) or "Fast Tracking" (performing Design and Development tasks in parallel) will be considered.

7. COST MANAGEMENT PLAN

7.1 Introduction to Cost Management

The Cost Management Plan for the UEMS project defines the processes for estimating, budgeting, and controlling costs so that the project can be completed within the approved budget. Since the university has a fixed allocation for IT digital transformation, maintaining financial discipline is crucial. This plan ensures that every Saudi Riyal (SAR) spent is aligned with the project's strategic value.

7.2 Cost Estimation

We have utilized **Bottom-Up Estimation** by calculating the costs of individual work packages from the WBS. The estimations include labor, software licenses, hardware infrastructure, and testing environments.

Resource Category	Description	Estimated Cost (SAR)
Human Resources (Labor)	Project Manager, 2 Developers, 1 UI/UX Designer, 1 QA Tester	120,000
Software & Licenses	Cloud Hosting (Azure/AWS), Security Certificates (SSL), Database License	15,000
Hardware & Infrastructure	Local backup servers and network optimization	10,000
Testing & Quality	External security audit and performance testing tools	5,000
Contingency Reserve	10% for unforeseen risks (e.g., scope changes)	15,000
Total Estimated Cost		**165,000 SAR**

7.3 Budget Breakdown

The budget is distributed across the 10-week lifecycle to ensure liquidity and financial oversight.

- **Phase 1 (Initiation & Analysis):** 15% (Personnel costs for requirements gathering)
- **Phase 2 (System Design):** 20% (UI/UX design and architecture planning)
- **Phase 3 (Development/Sprints):** 45% (The most resource-intensive phase)
- **Phase 4 (Testing & QA):** 10% (Security audits and bug fixing)
- **Phase 5 (Closing):** 10% (Documentation and handover)

7.4 Cost Baseline

The Cost Baseline is the authorized time-phased budget. For UEMS, the S-Curve represents the cumulative spending over the 10 weeks. Spending starts slowly in the planning weeks, peaks during the 4-week Development Sprints, and tapers off during the Closing phase.

7.5 Cost Control Methods

To ensure the project stays within the 165,000 SAR limit, the following controls are implemented:

- **Earned Value Management (EVM):** We will track **Planned Value (PV)**, **Actual Cost (AC)**, and **Earned Value (EV)** weekly.
- **Variance Analysis:** Any Cost Variance (CV) or Schedule Variance (SV) exceeding 5% will require an immediate review meeting with the Project Sponsor.
- **Change Control Integration:** As defined in the Integration Plan, no additional costs can be incurred without an approved Change Request, ensuring that "Scope Creep" does not lead to "Budget Creep.

8. QUALITY MANAGEMENT PLAN

8.1 Introduction to Quality Management

Quality Management for the University Event Management System (UEMS) ensures that the final deliverable meets the predefined technical requirements and provides a seamless user experience for the university community. In IT projects, quality is not just about the absence of bugs, but also about usability, security, and performance. This plan outlines the standards and procedures used to ensure that the UEMS is reliable and fit for purpose.

8.2 Quality Standards and Regulations

To align with international best practices, the UEMS project will adhere to the following standards:

- **ISO/IEC 25010:** A standard for System and Software Quality Requirements and Evaluation (SQuaRE), focusing on functional suitability, performance efficiency, and security.
- **WCAG 2.1:** Web Content Accessibility Guidelines to ensure the portal is accessible to all university members, including those with disabilities.
- **University IT Security Policy:** Ensuring all data handling complies with local institutional security and privacy protocols.

8.3 Quality Assurance (QA) vs. Quality Control (QC)

A dual approach is used to maintain high standards throughout the 10-week lifecycle.

8.3.1 Quality Assurance (Process-Oriented)

QA focuses on the processes used to create the deliverables. We aim to prevent defects before they occur.

- **Code Reviews:** Peer reviews of backend logic and database queries during Sprints.
- **Standard Checklists:** Ensuring every module follows the university's UI/UX design system.
- **Documentation Standards:** Maintaining a clear WBS dictionary and technical documentation to prevent misunderstandings.

8.3.2 Quality Control (Product-Oriented)

QC focuses on identifying defects in the actual product through testing.

- **Unit Testing:** Testing individual components (e.g., the "Pay Now" button) to ensure they function in isolation.

- **Integration Testing:** Verifying that the registration module correctly updates the centralized event calendar.
- **Security Scanning:** Running automated tools to check for vulnerabilities such as SQL injection or cross-site scripting (XSS).

8.4 Key Performance Indicators (KPIs) and Quality Metrics

The following metrics will be used to measure the quality of the UEMS:

- **Response Time:** The system must load the event dashboard in under 2 seconds for 95% of users.
- **Bug Density:** Less than 0.5 critical bugs per 1,000 lines of code before UAT.
- **System Availability:** The portal must maintain an uptime of 99.9% during the testing and deployment phase.
- **User Satisfaction:** A score of at least 4/5 in the post-UAT survey from university staff.

8.5 Acceptance Criteria

The project is considered "high quality" and ready for handover only when:

1. All "Must-Have" requirements defined in the Scope Statement (Section 5.2) are fully functional.
2. The system passes all security and stress tests conducted by the IT Department.
3. Formal sign-off is received from the Project Sponsor following the User Acceptance Testing (UAT) in Week 9.

9. HUMAN RESOURCE (TEAM) MANAGEMENT PLAN

9.1 Introduction to HR Management

The Human Resource Management Plan for the UEMS project provides guidance on how team members will be defined, recruited, managed, and eventually released. Since this is an IT project within a university setting, the team consists of a mix of project management professionals and technical specialists. The goal is to ensure that the right people with the right skills are assigned to the correct tasks to meet the 10-week deadline.

9.2 Project Team Structure

The UEMS project follows a **Functional Project Organization** structure, where the project team operates under the oversight of the University IT Department.

- **Project Sponsor:** Director of University IT (Provides funding and high-level approval).
- **Project Manager (Ajmal Hamidi):** Responsible for overall project success, planning, and coordination.
- **Business Analyst:** Responsible for gathering requirements from university departments.
- **Technical Lead / Full-Stack Developers (2):** Responsible for backend logic, database, and frontend development.
- **UI/UX Designer:** Responsible for the visual interface and user experience.
- **Quality Assurance (QA) Specialist:** Responsible for testing and system security.

9.3 Roles and Responsibilities

Role	Responsibility	Required Skills
Project Manager	Planning, scheduling, risk management, and stakeholder communication.	PMBOK knowledge, Leadership, Communication.
Business Analyst	Translating university needs into technical specifications.	Analytical thinking, Requirements gathering.
Developers	Coding the registration, resource allocation, and reporting modules.	SQL, Web Frameworks (e.g., React/Node.js), Security.
UI/UX Designer	Creating wireframes and ensuring the portal is easy to navigate.	Adobe XD/Figma, User Psychology.
QA Specialist	Executing test plans and identifying system bugs.	Detail-oriented, Automated testing tools.

9.4 RACI Matrix

The RACI matrix (Responsible, Accountable, Consulted, Informed) defines the level of involvement for each stakeholder in key project activities.

Activity / Task	Project Manager	Developers	UI/UX Designer	IT Sponsor
Project Charter	A / R	I	I	C
Requirements Gathering	A	C	C	R
WBS & Schedule	A / R	C	I	I
Database Design	A	R	C	I
Coding Modules	A	R	C	I
Security Testing	A	C	I	R
Final Handover	A / R	I	I	C

Key: ***R*** *= Responsible,* ***A*** *= Accountable,* ***C*** *= Consulted,* ***I*** *= Informed.*

9.5 Team Development and Management

Given the short duration (10 weeks), the team must reach high productivity quickly.

- **Stages of Team Development:** We will move the team through the *Forming, Storming, Norming,* and *Performing* stages within the first two weeks through team-building meetings and clear role definitions.
- **Conflict Management:** Any technical disagreements between developers or designers will be resolved using a **Collaborative/Problem-Solving** approach to ensure the project schedule is not impacted.
- **Recognition and Rewards:** Team members who complete their Sprints ahead of schedule with zero critical bugs will be recognized in the weekly status reports to the Project Sponsor.

10. COMMUNICATION MANAGEMENT PLAN

10.1 Introduction to Communication Management

The Communication Management Plan for the University Event Management System (UEMS) identifies the information needs of the stakeholders and defines how that information will be collected, stored, and distributed. Effective communication is critical for UEMS to resolve the "Poor Communication" issues mentioned in the project brief. This plan ensures that the right information reaches the right person at the right time using the most effective channel.

10.2 Communication Stakeholder Requirements

We have categorized the stakeholders based on their information needs and preferred frequency of communication.

Stakeholder	Information Needed	Frequency	Method/Format
Project Sponsor	High-level progress, budget status, and major risks.	Monthly	Formal Status Report / Presentation
Project Team	Detailed task assignments, technical issues, and daily progress.	Daily	Daily Stand-ups / Slack or MS Teams
University IT Dept	Technical specifications, security protocols, and integration data.	Bi-weekly	Technical Review Meetings / Email
Event Organizers	Feature updates, UI feedback requests, and training schedules.	Weekly	Email / Prototype Demo Sessions
University Students	System launch date and registration instructions.	Milestone-based	Social Media / University Newsletter

10.3 Communication Tools and Technologies

To ensure seamless information flow, the following tools will be utilized:

- **Project Management Tool:** MS Project/Excel for schedule and WBS tracking.
- **Email:** For formal correspondence and official project announcements.
- **Collaboration Platform (MS Teams/Slack):** For real-time technical discussions and file sharing (e.g., sharing the Database Schema).
- **Virtual Meetings (Zoom/Teams):** For weekly synchronization and stakeholder feedback sessions.

- **Shared Repository (GitHub/SharePoint):** For version control of technical code and project documentation.

10.4 Meeting Schedule

The following meetings are established to maintain project momentum:

1. **Kick-off Meeting (Week 1):** To align all stakeholders on project goals and the Charter.
2. **Daily Stand-up (Internal Team):** 15-minute meeting to discuss what was done yesterday, what will be done today, and any "blockers."
3. **Weekly Status Meeting:** Reviewing the Gantt chart, budget, and addressing any new risks.
4. **Sprint Review (Every 2 weeks):** Demonstrating the current "increment" of the UEMS (e.g., the registration form) to the IT department for feedback.
5. **Project Closure Meeting (Week 10):** Final handover and review of "Lessons Learned."

10.5 Reporting Structure

- **Status Reports:** The Project Manager (Ajmal Hamidi) will issue a "Project Health Report" every Friday. This report uses a **RAG (Red-Amber-Green)** status to indicate if the schedule and budget are on track.
- **Technical Documentation:** Developers will update the technical log after every Sprint to ensure the University IT Department has a record of all system configurations.

10.6 Escalation Matrix

In the event of a conflict or a critical issue that cannot be resolved at the team level (e.g., a hardware procurement delay):

1. **Level 1:** Discussion between the Project Manager and the Technical Lead (Resolution within 24 hours).
2. **Level 2:** If unresolved, the issue is escalated to the **University IT Sponsor** (Resolution within 48 hours).
3. **Level 3:** For issues affecting the 10-week deadline, a formal meeting with **University Management** is required.

11. RISK MANAGEMENT PLAN

11.1 Introduction to Risk Management

Risk Management for the University Event Management System (UEMS) is a proactive process of identifying, analyzing, and responding to project risks. The goal is to maximize the probability and impact of positive events and minimize the probability and impact of adverse events. Given the 10-week constraint and the technical nature of the UEMS, early identification of potential "Scheduling Conflicts" and "Technical Failures" is essential for project success.

11.2 Risk Identification

We have identified 10 specific risks that could impact the UEMS project, categorized by their source (Technical, External, Organizational, and Project Management).

1. **System Integration Failure:** Inability to sync UEMS with the university's existing student/staff authentication database.
2. **Scope Creep:** University departments requesting additional features (e.g., live streaming) mid-project.
3. **Data Security Breach:** Unauthorized access to participant personal information or payment data.
4. **Resource Unavailability:** Key developers or the Project Manager being unavailable due to illness or other commitments.
5. **Server Downtime:** Technical failure of university servers during the critical testing or launch phase.
6. **Low User Adoption:** Faculty or administrative staff refusing to switch from manual processes to the digital system.
7. **Payment Gateway Delays:** Administrative or technical hurdles in approving the online payment integration.
8. **Inaccurate Cost Estimation:** Underestimating the cost of third-party security audits or cloud hosting.
9. **Schedule Slippage:** Delays in the Design phase impacting the start of the Development Sprints.
10. **Browser Incompatibility:** The system failing to function correctly on older browsers used by some university departments.

11.3 Risk Probability and Impact Matrix

To prioritize these risks, we use a Qualitative Risk Analysis matrix. Risks are scored from 1 to 5 for both Probability (likelihood) and Impact (severity).

Risk ID	Risk Description	Probability (1-5)	Impact (1-5)	Risk Score (PxI)	Priority
R1	Integration Failure	3	5	15	**High**
R2	Scope Creep	4	3	12	**Medium**
R3	Data Security Breach	2	5	10	**Medium**
R5	Server Downtime	2	5	10	**Medium**
R9	Schedule Slippage	3	4	12	**High**

11.4 Risk Register and Mitigation Strategies

The Risk Register below outlines the response strategies for the top identified risks.

Risk ID	Risk Event	Trigger	Response Strategy	Mitigation/Action Plan
R1	Integration Failure	Authentication errors in Week 2	**Mitigate**	Involve University IT DBAs in the Requirements phase (Week 2).
R2	Scope Creep	Requests for new features	**Avoid**	Enforce the formal Change Control process defined in Section 4.5.
R3	Data Breach	Failed security scan	**Mitigate**	Implement AES-256 encryption and conduct a mid-project security audit.
R6	Low Adoption	Negative feedback during UAT	**Mitigate**	Schedule training workshops and create "User Manuals" in Week 10.
R7	Payment Delay	Approval takes >2 weeks	**Transfer**	Use a pre-approved university payment provider to bypass new approvals.
R9	Schedule Slippage	Task delay >2 days	**Mitigate**	Use "Crashing" techniques (extra hours) to bring the Critical Path back on track.

11.5 Risk Monitoring and Control

Risk management is not a one-time activity.

- **Risk Audits:** During every weekly status meeting, the Risk Register will be reviewed to close outdated risks and identify new ones.
- **Contingency Reserves:** A 15,000 SAR budget (defined in Section 7.2) is reserved specifically to fund the mitigation actions for identified risks.

12. PROCUREMENT MANAGEMENT PLAN

12.1 Introduction to Procurement Management

The Procurement Management Plan for the University Event Management System (UEMS) describes how the project team will acquire goods and services from outside the performing organization. For an IT project of this scale, procurement involves making decisions about whether to develop certain components internally or purchase existing third-party solutions (e.g., cloud hosting, payment gateways, or security certificates).

12.2 Make-or-Buy Analysis

A "Make-or-Buy" analysis was conducted to determine which parts of the UEMS should be developed by the internal university project team and which should be procured externally.

Component	Decision	Justification
Core Application Logic	**Make**	Developing the core registration and resource logic internally ensures full customization to university policies and avoids long-term licensing fees.
Cloud Infrastructure	**Buy**	Purchasing cloud hosting (e.g., AWS or Azure) is more cost-effective and reliable than maintaining physical on-premise servers for a 10-week project.
Payment Gateway	**Buy**	Integrating an established API (like Stripe or a local bank gateway) ensures high security and compliance without the massive overhead of building a financial system.
Security Audit	**Buy**	Hiring an external specialized firm to perform penetration testing provides an unbiased and professional validation of the system's integrity.

12.3 Vendor Selection Criteria

When selecting external vendors for cloud services or security audits, the project team will use the following weighted criteria to ensure the best value:

1. **Technical Capability (40%):** Does the vendor have experience with university-scale IT systems?
2. **Cost (25%):** Is the pricing competitive and within the budget defined in Section 7?
3. **Security and Compliance (20%):** Does the vendor meet local and international data protection standards?

4. **Support and Reliability (15%):** What are the Service Level Agreements (SLAs) regarding uptime and response time?

12.4 Contract Types

The UEMS project will utilize two primary types of contracts to manage procurement risk:

- **Firm-Fixed-Price (FFP) Contracts:** Used for the external security audit. This protects the university from cost overruns, as the scope and price are agreed upon upfront.
- **Subscription-Based (SaaS) Agreements:** Used for cloud hosting and SMS notification services. This allows for scalability and ensures the university only pays for the resources consumed during the 10-week lifecycle.

12.5 Procurement Process

The procurement process follows these four steps:

1. **Plan Procurements:** Identifying the needs (Cloud, Security, API) and creating Statements of Work (SOW).
2. **Conduct Procurements:** Requesting quotes from approved vendors and selecting the best fit based on the criteria in 12.3.
3. **Control Procurements:** Monitoring vendor performance against the contract (e.g., ensuring 99.9% cloud uptime).
4. **Close Procurements:** Verifying that all services were rendered correctly and making final payments.

13. STAKEHOLDER MANAGEMENT PLAN

13.1 Introduction to Stakeholder Management

The Stakeholder Management Plan for the University Event Management System (UEMS) identifies the individuals, groups, or organizations that could impact or be impacted by the project. In a university setting, stakeholders have varying levels of influence and interest. This plan ensures that all stakeholders are engaged appropriately throughout the 10-week lifecycle to ensure system adoption and project success.

13.2 Stakeholder Register

The following register identifies the key stakeholders involved in the UEMS project.

Name/Group	Role in Project	Expectations	Influence	Interest
Dr. Salman (IT Dept)	Project Sponsor	On-time delivery, security compliance, and budget adherence.	High	High
Ajmal Hamidi	Project Manager	Successful project completion and academic excellence.	High	High
Faculty Organizers	Key Users	Ease of use, reliable room booking, and automated reporting.	Medium	High
University IT Staff	Technical Support	System stability, clean code, and easy maintenance.	Medium	Medium
Student Body	End Users	Simple registration process and mobile compatibility.	Low	High
Finance Department	Oversight	Transparent tracking of event-related payments.	High	Low

13.3 Power-Interest Grid

To determine the best engagement strategy, stakeholders are mapped onto a Power-Interest Grid. This allows the Project Manager to prioritize efforts.

- **High Power / High Interest (Manage Closely):** Project Sponsor and Project Manager. These individuals require the most attention and frequent communication.
- **High Power / Low Interest (Keep Satisfied):** Finance Department. They need to be kept informed of budget status but do not need daily technical updates.

- **Low Power / High Interest (Keep Informed):** Faculty Organizers and Students. They are the primary users and need to feel involved to ensure the system is adopted.
- **Low Power / Low Interest (Monitor):** External vendors or general university staff who are not directly involved in event management.

13.4 Stakeholder Engagement Strategy

Based on the grid, the following engagement strategies will be applied:

Stakeholder Group	Engagement Approach	Key Message
Manage Closely	Weekly face-to-face or virtual meetings and formal status reports.	"The project is on track and meeting all university IT standards."
Keep Satisfied	Monthly financial updates and milestone approvals.	"The project is adhering to the allocated 165,000 SAR budget."
Keep Informed	Prototype demonstrations (Sprints) and user feedback surveys.	"Your feedback is being used to make the system easier for you to use."
Monitor	General announcements through the university portal or newsletter.	"UEMS is coming soon to modernize our event management."

13.5 Stakeholder Engagement Assessment Matrix

We will monitor the "Current" (C) versus "Desired" (D) engagement level of key stakeholders:

Stakeholder	Unaware	Resistant	Neutral	Supportive	Leading
Project Sponsor				C	D
Faculty Organizers		C		D	
Technical Team				C	D

Strategy: For "Resistant" faculty members, we will conduct special demo sessions in Week 8 to show how UEMS reduces their manual workload, moving them to a "Supportive" state.

14. CONCLUSION AND FINAL SUMMARY

14.1 Project Synthesis

The Project Management Plan for the University Event Management System (UEMS) provides a structured and professional roadmap for transitioning the university's event coordination from a manual, error-prone process to a centralized digital ecosystem. By meticulously applying the 10 Knowledge Areas of the PMBOK guide over a simulated 10-week lifecycle, this plan ensures that all technical, financial, and operational constraints are addressed before a single line of code is written.

14.2 Achievement of Objectives

Through the development of this comprehensive plan, the following project goals have been secured:

- **Integration & Scope:** A clear boundary has been established through the Project Charter and WBS, ensuring that "Scope Creep" is managed and the project remains focused on core university needs.
- **Timeline & Budget:** By defining a 10-week Gantt chart and a detailed 165,000 SAR budget, the project demonstrates financial and temporal feasibility.
- **Risk & Quality:** The identification of 10 critical risks and the establishment of a dual QA/QC approach provide a safety net that guarantees system reliability and data security.
- **Stakeholder Synergy:** The Communication and Stakeholder Management plans ensure that the university administration, faculty, and students are aligned and engaged, which is vital for long-term system adoption.

14.3 Final Recommendation

The UEMS project is not merely a software development initiative; it is a strategic step toward the university's digital transformation. The selection of the Agile (Scrum) methodology allows for the flexibility required in an academic environment while maintaining the discipline of traditional project management.

Based on the detailed analysis provided in this report, the UEMS project is deemed viable and ready for the execution phase. Upon implementation, the university can expect a 100% reduction in venue scheduling conflicts, significantly improved financial transparency, and a superior experience for all event participants.

14.4 Closing Statement

As the Project Manager, I, Ajmal Hamidi, submit this Final Project Management Plan as a complete framework for the UEMS. This document serves as the official baseline for the project and a testament to the rigorous application of IT project management principles.